

FURONG JIA

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Durham, North Carolina, United States

RESEARCH STATEMENT

I am broadly interested in making AI systems, especially large language models, **reliable, interpretable, and robust** by understanding *how data shapes model behavior at training and inference time*. My interests include methods that leverage properties of diverse data (text, time-series, multimodal) to make models adaptable to changes that arise during both training and deployment. A parallel interest is applying these approaches in healthcare, where robustness, generalizability, and responsible deployment are critical.

RESEARCH INTEREST

Natural Language Processing, Data-centric Machine Learning, Healthcare, Time Series, Multimodal Machine Learning

EDUCATION

• Duke University

Doctor of Philosophy in Computer Science

◦ Advisor: Monica Agrawal

Aug 2024 - Present

Durham, NC

GPA: 4.0/4.0

• University of Southern California

Bachelor of Science in Computer Science & Bachelor of Science in Applied and Computational Mathematics

◦ W.V.T. Rusch Engineering Honors Program, Viterbi Grand Challenges Scholar

Aug 2020 - May 2024

Los Angeles, CA

GPA: 3.96/4.0

PUBLICATIONS

Conference:

[1] Interpreting Dataset Shift in Clinical Notes

ML4H 2025 (Machine Learning for Health Symposium)

Shariar Vaez-Ghaemi*, **Furong Jia***, Monica Agrawal.

[2] Diagnosing our datasets: How does my language model understand clinical text?

CHIL 2025 (7th Annual Conference on Health, Inference, and Learning)

Furong Jia, David Sontag, Monica Agrawal.

[3] GPT4MTS: Prompt-based Large Language Model for Multimodal Time-series Forecasting.

EAAI-24 (The 14th Symposium on Educational Advances in Artificial Intelligence)

Furong Jia, Kevin Wang, Yixiang Zheng, Defu Cao, Yan Liu. (2024).

[4] TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting.

ICLR 2024 (The Twelfth International Conference on Learning Representations)

Defu Cao, **Furong Jia**, Serkan O Arik, Tomas Pfister, Yixiang Zheng, Wen Ye, Yan Liu.

[5] I2I: Initializing Adapters with Improvised Knowledge.

CoLLAs 2023 (Second Conference on Lifelong Learning Agents)

Tejas Srinivasan, **Furong Jia**, Mohammad Rostami, Jesse Thomason.

Workshop/Findings:

[6] Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM

ML4H 2025 Findings (Machine Learning for Health Symposium)

Furong Jia*, Yuan Pu*, Finn Guo, Monica Agrawal.

[7] What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking

ML4H 2025 Findings (Machine Learning for Health Symposium)

Raymond M Xiong, **Furong Jia**, Lionel Wong, Monica Agrawal.

Under Review:

[8] Batch-of-Thought: Cross-Instance Learning for Enhanced LLM Reasoning

Xuan Yang, **Furong Jia**, Roy Xie, Xi Xiong, Jian Li, Monica Agrawal

[9] How Faithfully do Language Models Acquire Probabilistic Knowledge?

Furong Jia, Xunjian Yin, Yuan Pu, Monica Agrawal

RESEARCH EXPERIENCE

• IBM Research, Core AI Team

Research Intern | Mentor: Akash Srivastava

May 2026 – Aug 2026

Boston, MA

- Developing modular AI-agent harnesses for scientific discovery, comparing orchestration, structured search, memory, and role-based components under matched backbone, interface, and compute budgets.
- Studying where humans should intervene in agentic scientific workflows, with focus on failure modes, task boundaries beyond current agent capability, and help-seeking policies that elicit useful human input.

• DukeNLP, Duke University

Advisor: Monica Agrawal

Aug 2024 - Present

Durham, NC

- **Understanding LLM dynamics as a function of pretraining data**
 - * We examined how evidence in pretraining corpora shapes clinical LLM behavior. We linked keyword-pair occurrences in pretraining data to model performance on clinical jargon understanding and unsupported medical claims, curated *MedLingo* from real-world clinical-note abbreviations, and analyzed sources of clinical information in pretraining datasets to guide future corpus design. [2]
 - * We tested how much clinical diagnostic benchmark performance can be explained by explicit corpus-frequency signals rather than LLM inference alone. We introduced *Frequency-Based Probabilistic Ranker (FBPR)*, a lightweight Naive Bayes method that scores diagnoses using clinical-concept co-occurrence statistics from the same pretraining corpora as the evaluated models. FBPR achieved performance comparable to corpus-aligned LLMs, answered largely different subsets of questions, and improved performance when combined with LLM predictions through a simple hybrid method. These results show that explicit probabilistic baselines can serve as meaningful, historically grounded reference points and complementary signals for hybridization. [6]
 - * We studied how faithfully LLMs acquire probabilistic knowledge from pretraining text. We built a controlled continual-pretraining framework where known data-generating networks define ground-truth distributions, synthetic clinical-style corpora verbalize sampled configurations, and OLMo-2 models are evaluated by comparing native next-token probabilities with ground-truth posteriors. We found that models recover simple conditional distributions well but show scale-dependent recovery ceilings in larger settings, and that they often learn the direction of probability shifts while underestimating their magnitude. [9]
- **Interpreting Dataset Shift in Clinical Notes**
 - * We addressed the need to explain, rather than only detect, distribution shifts in clinical text. We established an extensible benchmark suite that induces synthetic shifts using real clinical notes, developed methods for evaluating generated shift explanations, and introduced *SIReNs*, an end-to-end approach that explains distributional differences by selecting representative notes from each distribution. We are also extending this work to benchmark agent capabilities and agent harnesses for interpreting shifts in multi-factorial settings and more general textual domains. [1]

• Melady Lab, University of Southern California

Advisor: Yan Liu

May 2023 - June 2024

Los Angeles, CA

- **Large Language Models for Time Series Forecasting**
 - * We introduced *GPT4MTS*, a prompt-based framework that fuses numeric time series with texts. We also built a GDELT-derived multimodal news impact dataset and showed consistent forecasting gains over strong unimodal baselines, demonstrating effective multimodal fusion and the value of extra-textual information. [3]
 - * We developed *TEMPO*, a GPT architecture-based time series framework that combines seasonal trend decomposition with adaptive, selection-based prompting for non-stationary distribution shift. It achieves zero-shot state-of-the-art across diverse and unseen (including multi-modal) benchmarks, indicating strong potential as a foundation model for time series forecasting. [4]

• GLAMOR Lab, University of Southern California

Advisor: Jesse Thomason

Aug 2022 - Present

Los Angeles, CA

- **Initializing Adapters with Improvised Knowledge for Multimodal Continual Learning**
 - * We applied parameter-efficient knowledge transfer using adapters in multimodal models for visual question-answering tasks. We implemented and trained model architectures on multiple vision-language tasks, including GQA, COCO-QA, and AQUA, to enhance forward transfer and reduce knowledge forgetting. [5]

HONORS AND AWARDS

- **University of Southern California:** Albert Dorman Future Leader Award (2024), Student Recognition Awards (2024), Provost's Research Fellowship (Fall 2023), CURVE Research Fellowship (2022-2023), Academic Achievement Award (2021), ABC Innovation Prize (2021).

SERVICE

- **Reviewer:** NeurIPS '26, NeurIPS '25, ML4H '25, ACL ARR 2024 (June, October), ACL ARR 2025 (February, May).
- **Teaching Assistant:** Natural Language Processing (Duke, Spring '26); Applied Machine Learning (Duke, Spring '25); Algorithms and Theory of Computing (USC, Spring '22); Web Publishing and Front-end Development (USC, Fall '21 - Spring '24).